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END SEMESTER EXAMINATION – 2024

Name of the Course: B. Tech

Semester VII

Name of the Paper: Artificial Intelligence and Pattern Recognition Paper Code: TOE 713

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions from a, b & c in each main question
- (iii) Total for each main question is 20 marks and each sub-question is 10 marks.

Q1	(20 marks)	
(a)	By using functional blocks clearly describe Pattern classification. What are the steps in design of binary classifier that separates Orange vs. Watermelon using size and weight features.	CO5
(b)	Give at least 4 different definitions of Artificial Intelligence. Also, cite 4 applications of AI that involves Pattern Recognition in the real-world and discuss its statistical performance.	CO3
(c)	Describe giving 2 examples each of the terms: Data, Information, Knowledge, Language.	CO2
Q2	(20 marks)	
(a)	Explain various Knowledge representation methods and Knowledge types in AI. Also, using functional blocks describe a generic AI engine under Production type.	CO3
(b)	Explain overfitting vs. underfitting in classifier design. Given the following 2- dimensional dataset with weight : = W (numeric), size : = Z (small, medium, large) variables denoting Apples and Kiwi categories, use the k-NN for k = 1 and k = 4 to perform classification. Two Inputs given to classify are – (8.5, m); (11.75, l) Apples = {(6, s); (7, s); (8, m); (10, m); (13, l), (17, l)} Kiwi = { (3, s); (4, s); (7, m); (8, m); (11, l); (14, l)}	CO6
(c)	Explain Clustering giving basic steps to perform – hierarchical and partitional algorithms. What help does PCA provide while performing Clustering.	CO4
Q3	(20 marks)	
(a)	Explain Binary hypothesis testing and depict the type – I and type - II errors. Also, derive Log Likelihood Rule (LLR) test for classification.	CO4
(b)	Given: $X_i = 1, 3, 5, \dots, 9, 15, 18, 28, 33$ and $Y_i = 1, 5, 8, 12, 15, 21, 35, 41, 52$ is 2-dimensional feature space that pairwise jointly denotes the data samples. Perform K-means clustering with 2 seed values given by (7, 11) and (16, 25). How will clustering perform under missing sample as seen in time series for X.	CO6
(c)	Explain syntactic pattern recognition with an example, preferably for natural language analysis.	CO3
Q4	(20 marks)	
(a)	Giving steps clearly explain basic Perceptron, how does it perform learning. Show three different activation functions.	CO2
(b)	Explain Bias and Variance for data model. How can it be minimized. How does the bias – variance effect ANN.	CO3
(c)	State and prove the Bayes theorem, show DFS and BFS search for Bayesian network.	CO1
Q5	(20 marks)	
(a)	A binary file data such as 1010111100001 is sent on noisy channel with error probability = 0.03, assuming fair playing conditions compute probability of <u>exactly</u> 6 errors ? and <u>at most</u> 6 errors ?	CO2
(b)	In exploratory analytic, what is Scatter and explain measure for correlation showing its various geometries also. Given $Y = X - 5$, compute mean and variance of Y if the values for X are 3 and 4.	CO4
(c)	Explain the following: feature engineering in CNN, backpropagation, advantages of ReLU type activation, Types of AI search.	CO2

